

Unit IV

DIVISIBILITY THEORY AND CANONICAL DECOMPOSITION

Course Objective – Cob4: To explain concepts of number theory and its applications.

Course Outcome – CO4: Apply divisibility theory and canonical decompositions in problems.

Bloom's Taxonomy Level: K1, K2, K3, K4

Part A

S.No.	Question	BTL*
1.	Find the quotient and the remainder when the first integer is divided by the second using division algorithm (i). 75, 28 (ii). 57, 75 (iii). -25, 3 (iv). -23, 25. (i). $75 = 2(28) + 19$ (ii). $57 = 0(75) + 57$ (iii). $-25 = (-9)(3) + 2$ (iv). $-23 = (-1)(25) + 2$	K2
2.	If a and b be positive integers such that $a b$ and $b a$, then Prove that $a = b$. Given that both a and b are positive (i.e) $a > 0, b > 0 \dots \dots (1)$ And also $a b$ & $b a \Rightarrow b = q_1a \dots (2)$ & $a = q_2b \dots (3)$ where $q_1, q_2 \in \mathbb{Z}^+$ Now substitute (3) on (2), we get $b = q_1[q_2b] = q_1q_2b$ $\Rightarrow b - q_1q_2b = 0$ $\Rightarrow b(1 - q_1q_2) = 0$ $\Rightarrow 1 - q_1q_2 = 0$, since $b > 0$ $\Rightarrow q_1q_2 = 1$, with $q_1, q_2 \in \mathbb{Z}^+$ $\Rightarrow q_1 = 1$ & $q_2 = 1$ substitute above values in (2) & (3), we get $a = b$.	K2
3.	Define Floor function and write the application of Floor function in number theory. The floor function of x denoted by $\lfloor x \rfloor$ is defined as the greatest integer $\leq x$. <u>Application of Floor function.</u> Let a and b be any two positive integers. Then the number of positive integers $\leq a$ and divisible by b is $\lfloor a/b \rfloor$.	K1
4.	Find the number of positive integers ≤ 3076 and (i). Divisible by 19 (ii). not divisible by 17. (i). $\left. \begin{array}{l} \text{No. of positive integers } \leq 3076 \\ \text{and divisible by 19} \end{array} \right\} = \lfloor 3076/19 \rfloor = \lfloor 161.89 \rfloor = 161$ (ii). $\left. \begin{array}{l} \text{No. of positive integers } \leq 3076 \\ \text{and divisible by 17} \end{array} \right\} = \lfloor 3076/17 \rfloor = \lfloor 180.94 \rfloor = 180$ $\left. \begin{array}{l} \text{No. of positive integers } \leq 3076 \\ \text{and not divisible by 17} \end{array} \right\} = 3076 - 180 = 2896$	K2

5.	Prove each by cases, when n is an arbitrary integer: (a) $(n^2 + n)$ is an even integer. (b) $(2n^3 + 3n^2 + n)$ is an even integer. (a). Given that $(n^2 + n) = n(n + 1) \dots \dots (1)$ Case1. Suppose n is even, the $n + 1$ is odd By property “Product of even and odd integers is even” $(n^2 + n) = n(n + 1)$ is Even Case2. Suppose n is odd, the $n + 1$ is even By property “Product of even and odd integers is even” $(n^2 + n) = n(n + 1)$ is Even (b). Given that $(n^2 + n) = n(n + 1) \dots \dots (1)$ Case1. Suppose n is even, the $n + 1$ is odd By property “Product of even and odd integers is even” $(n^2 + n) = n(n + 1)$ is Even Case2. Suppose n is odd, the $n + 1$ is even By property “Product of even and odd integers is even” $(n^2 + n) = n(n + 1)$ is Even	K2
6.	Define Relatively prime with an example. Relatively prime: If $(a, b) = 1$, then the integers a and b are said to be relatively prime. Example: Every two consecutive integers are relatively prime. (i.e) $(8, 9) = 1$, $(15, 16) = 1$	K1
7.	Write number-theoretic function. Let $p_1, p_2, \dots, p_t$ be primes $\leq \sqrt{n}$. Then $\pi(n) = (n - 1) + \pi(\sqrt{n}) - \sum_i \left\lfloor \frac{n}{p_i} \right\rfloor + \sum_{i < j} \left\lfloor \frac{n}{p_i p_j} \right\rfloor - \sum_{i < j < k} \left\lfloor \frac{n}{p_i p_j p_k} \right\rfloor - \dots (-1)^t \left\lfloor \frac{n}{p_i p_j \dots p_t} \right\rfloor$	K1
8.	State fundamental theorem of arithmetic. Every integer n (≥ 2) either is a prime or can be expressed as a product of primes. The factorization in to primes is unique expect for the order of the factors.	K1
9.	State necessary and sufficient condition that LDE $ax + by = c$ is solvable and express their general solutions. The LDE $ax + by = c$ is solvable if and only if $d c$, where $d = (a, b)$. General solutions If x_0, y_0 is initial solution of LDE, then all its solutions are given by $x = x_0 + \left(\frac{b}{d}\right)t, \quad y = y_0 - \left(\frac{a}{d}\right)t$ Where t is arbitrary integer.	K1
10.	Determine whether the LDEs $12x + 8y = 30$, $2x + 3y = 4$ and $6x + 8y = 25$ are solvable. (i). $(12, 8) = 4$ and “4 doesn’t divide 30” $\Rightarrow 12x + 8y = 30$ cannot be solved (ii). $(2, 3) = 1$ and “1 4” $\Rightarrow 2x + 3y = 4$ is solvable (iii). $(6, 8) = 2$ and “2 doesn’t divide 25” $\Rightarrow 6x + 8y = 25$ cannot be solved.	K2

Part B

S.No.	Question	BTL *
1.	State division algorithm. Let a, b, c, α and β be any integers. Then Prove or disprove (a) If $a b$ and $b c$, then $a c$. (b) If $a b$ and $a c$, then $a (\alpha b + \beta c)$ (c) If $a b$, then $a bc$ (d) If $a (b + c)$, then $a b$ and $a c$. (e) If $a bc$, then $a b$ and $a c$.	K4
2.	If a and b be any positive integers, then Show that the number of positive integers $\leq a$ and divisible by b is $\lfloor a/b \rfloor$.	K4
3.	(i). Find the number of positive integers after 1947 through 3347 that are divisible by 11 or 17. (ii). Find the number of positive integers after 1976 through 3776 that are divisible by neither 13 nor 19. (iii). Show that the number leap years L after 1800 and not exceeding a given year y is given by, $L = \lfloor y/4 \rfloor - \lfloor y/100 \rfloor + \lfloor y/400 \rfloor - 436$.	K4
4.	Prove that $30 \mid (n^5 - n)$, when n is an arbitrary integer.	K4
5.	(i). Prove every integer $n \geq 2$ has a prime factor (ii). Prove that there are infinitely many primes.	K4
6.	Prove that every composite $n \geq 2$ has a prime factor $\leq \lfloor \sqrt{n} \rfloor$ and determine whether the integers (a). 1601 (b) 667 (c) 601 (d) 527 are prime or not	K4
7.	Determine the following (a). Find the number of primes ≤ 100 . (b). Find $\pi(125)$.	K4
8.	Prove that for every positive integer n , there are n consecutive integers that are composite numbers and find six consecutive integers that are composite.	K4
9.	Any two consecutive Fibonacci numbers are relatively prime.	K4
10.	Prove that GCD of two positive integers a and b is a linear combination of a and b .	K4
11.	Apply the Euclidean algorithm to evaluate (a). (2076, 1076) and express (2076, 1076) as linear combination of 2076 and 1076 (b). (3076, 1776) and express (3076, 1776) as linear combination of 3076 and 1776.	K4
12.	(i). Using recursion, evaluate (a). (14, 18, 21, 36, 48). (b) $(a^2b^2, ab^3, a^2b^3, a^3b^4, a^4b^4)$ (ii). Using the canonical decomposition of (a). 1050 and 2574 (b). 294 and 450 and find the GCD and LCM.	K4
13.	(i). Eleven travellers entered the beautiful forest. They found 17 equal heaps of plantains and 5 single fruits and divided them equally. Find the number of fruits in each heap. (ii). If a sheep is worth 7 Rs, a hen 5 Rs and 3 chicks together a rupee. How many sheeps, hens and chicks, totaling 100 can be bought for 100 rupees?	K4